

Optimization Methods for Machine Learning and Engineering

Errata

May 11, 2021

- This document collects errata for the version of the slides that was video-recorded for the winter term 2020/21.
- Corrected versions of the slides are made available as well.
- Check the date of the errata document and possible update dates on the title slide of each lecture.

1 Introduction, Convexity & Gradient Descent

Last updated on *Mai 11, 2021*.

Slide 17 Updated figure on 1st degree Taylor Expansions.

Slide 18 Added missing $\frac{1}{2}$ factor for the 2nd degree Taylor Expansion. Added the condition of continuous second partial derivatives for Schwarz's Theorem.

Slide 22 The subgradient is the direction of a tangent line below the function, not the tangent line itself.

Old Version

We call the tangent line that touches f at $\mathbf{x}$ the subgradient at $\mathbf{x}$

$$\partial f_{\mathbf{x}}(\mathbf{z}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{z} - \mathbf{x}), \quad \forall \mathbf{x}, \mathbf{z} \in X$$

- f is convex over X iff $f(\mathbf{z}) \geq \partial f_{\mathbf{x}}(\mathbf{z})$, $\forall \mathbf{x}, \mathbf{z} \in X$
- f is strictly convex over X iff the above inequality is strict whenever $\mathbf{x} \neq \mathbf{z}$

What if f is not everywhere differentiable?

- A convex function must be continuous but is not necessarily differentiable at every point.
- If f is not differentiable, then $\partial f_{\mathbf{x}}$ is seen as the set of all subgradients that “lie below f and touch f at $\mathbf{x}$ ”.

Corrected Version

A subgradient of f at $\mathbf{x} \in X$ is a vector $\mathbf{g} \in \mathbb{R}^n$ for which

$$f(\mathbf{z}) \geq f(\mathbf{x}) + \mathbf{g}^\top (\mathbf{z} - \mathbf{x}), \quad \forall \mathbf{x}, \mathbf{z} \in X. \quad (1)$$

If f is convex and differentiable, then the gradient $\nabla f(\mathbf{x})$ is the only subgradient. For strictly convex f , the above inequality is strict whenever $\mathbf{x} \neq \mathbf{z}$.

What if f is not everywhere differentiable?

- A convex function must be continuous, but is not necessarily differentiable at every point.
- For convex functions, multiple subgradients exist for $\mathbf{x}$ where f is non-differentiable.
- The subdifferential $\partial f(\mathbf{x})$ is the set of all subgradients $\mathbf{g} \in \partial f(\mathbf{x})$ that hold Equation 1.

Slide 28 For clarity, show the gradient instead of the subdifferential in the figure (both are correct).

2 Newton Method, Loss Functions and Regularization

Last updated on *Mai 11, 2021*.

Slide 4 Unless indicated otherwise in the notation, all vectors are *column-vectors* in this course.

Slide 14 Corrected a typo in the Newton Method.

$$y = x - \frac{f(x)}{\nabla f(x)}$$

Slide 19 Logarithm is monotonically increasing, therefore preserves the location of the global minimum.

Slide 31 ... with fixed values $\mu > 0, q \geq 1$

3 Inequality Constraints

Last updated on *Mai 11, 2021*.

Slide 3 The correct term is to speak of a “sequence” instead of a “series”. A series would occur if we were summing the elements of a sequence.

$$B_n = [\frac{1}{n}, 1 - \frac{1}{n}], \quad \bigcup_{n=2}^{\infty} B_n = (0, 1)$$

This union of infinitely many closed sets is open!

Slide 12 With $x_{\text{tv}}, x_{\text{paper}}$ the number of ad runs, the number of people who saw the ad at least once is. . .

Slide 16 The notation for Stigler’s Diet was disambiguated from the general setting (α, β instead of $\mathbf{a}, \mathbf{b}$).
Use the greater-equal inequality constraint $x_j \geq 0$.

4 Equality Constraints

5 Optimization Applications

Last updated on *January 12, 2021*.

Slide 25 The variance of the mean estimator is $\sigma_e^2 = (\frac{1}{k})^2 \sigma_s^2 = \frac{1}{k} \sigma_a^2$.

6 Automatic Differentiation

Last updated on *January 11, 2021*.

Slide 7 Scalar Multiplication: $a(x + \dot{x}\varepsilon) = ax + a\dot{x}\varepsilon$

7 Optimization in Vector Spaces

Last updated on *February 12, 2021*.

Slide 9 The p -norms require $p \geq 1$.

8 Duality

Last updated on *February 04, 2021*.

Slide 9 Slater's Theorem uses the relative interior.

Slide 13 Make clear what is a “thought-experiment” for the exposition that is not entirely rigorous.
Correct an x instead of x .

Slide 14 Correctly use g for the vector of all inequality constraints. Correctly use x^* when an assertion shall only hold for the optimizer.

9 SVM

Last updated on *February 18, 2021*.

Slide 18 Let $p \in \mathbb{N}_+$ with $\mathbb{N}_+$ the positive natural numbers $1, 2, \dots$

$$G = \left\{ \sum_{i=1}^p \alpha_i k(\mathbf{y}_i, \cdot) \mid p \in \mathbb{N}_+, \alpha_i \in \mathbb{R}, \mathbf{y}_i \in X \right\}$$

10 Conic Programming

Last updated on *February 2, 2021*.

Slide 3 A proper cone is **convex**, pointed, closed and has a nonempty interior.

Slide 7 In the optimization problems, the total variation is penalized for x , not y .

Slide 12 Added clarification: The dual of a proper cone is itself proper.

Slide 16 Motivate the log-barrier for the Semi-Definite Cone.